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Cross-attention multi branch for Vietnamese sign language recognition: CrossViViT

Minh Hoang Chu[1], Hoang Diep Nguyen[1], Thi Ngoc Anh Nguyen[2] and Hoai Nam Vu[1,3*]

*Correspondence:
Hoai Nam Vu
namvh@ptit.edu.vn
[1]Young Innovation Research Laboratory on Digital Technology (YIRLoDT), Posts and Telecommunications Institute of Technology , Hanoi, Vietnam
[2]The University of Danang - University of Science and Education, Da Nang, Vietnam
[3]Faculty of Artificial Intelligence, Posts and Telecommunications Institute of Technology, Hanoi, Vietnam

Abstract
Sign language serves as the primary communication medium for individuals who are deaf or hard of hearing. Despite its critical importance, barriers persist in communication between the deaf community and the broader society, primarily due to limited sign language proficiency among the general population. While automated sign language recognition (ASLR) systems leveraging machine learning technologies offer a promising solution, existing approaches face challenges in optimizing the trade-off between computational efficiency and recognition accuracy. This study presents CrossViViT, a novel architecture that integrates cross-attention mechanisms with video vision Transformer networks to address these limitations. Drawing inspiration from multi-branch network architectures that combine diverse feature perspectives for flexible image recognition, our approach achieves both computational efficiency and high accuracy. The proposed model demonstrates exceptional performance on the Vietnamese Sign Language (VSL) dataset, achieving 92.47% accuracy in recognizing 50 distinct gestures across 8510 videos while maintaining computational efficiency at approximately 629 FLOPS.

Keywords Vietnamese sign recognition, Machine learning, Vision transformer, Cross-attention multi-branch

1 Introduction

Human activity recognition (HAR) is a specialized field of study focused on identifying specific actions performed by individuals through the analysis of various data sources [1]. These actions, referred to as human activities (HA), involve coordinated movements of body parts to accomplish a series of tasks. Human activities encompass various behaviors, including everyday tasks, sports activities, work-related tasks, and communication actions. The current research on human activity recognition predominantly employs two main methods: computer vision and sensor-based approaches [2]. Computer vision involves analyzing video data from digital cameras situated within the activity environment [3]. Conversely, sensor-based approaches utilize data from inertial sensors in devices such as smartwatches and mobile phones to identify human activities [4]. It is important to note that each approach is not without its limitations. Computer vision methods are susceptible to environmental variables such as lighting and

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obstructions. On the other hand, sensor-based methods encounter challenges in differentiating between similar activities and often require specialized sensors to address this issue [5]. Furthermore, the annotation of inertial sensor data is a costly and time-consuming, which impedes the establishment of large-scale datasets required for training effective recognition models [6].

Sign language recognition (SLR) is an essential branch of human action recognition. SLR models are used to simplify communication within the deaf community. In sign language conversation, human activities serve as a crucial means of communication for individuals with hearing and speech impairments, as they employ upper body movements to express vocabulary words. Moreover, sign language extends beyond human-to-human communication, facilitating communication between humans and machines [7]. However, one of the crucial requirements for SLR is accuracy, as the number of classes to be recognized in sign language can reach thousands. Video-based action recognition models often achieve higher accuracy than wearable sensor-based models, thus they are commonly used to build SLR applications in practice.

Significant advancements in video-based sign language recognition have been made in recent years, attributed to breakthroughs in deep learning and neural networks. The authors of [8] proposed a hybrid model combining Convolutional Neural Networks (CNNs) with Hidden Markov Models (HMMs) for continuous sign language recognition. CNNs extract spatial features from videos, while HMMs model temporal dependencies between signs. [9] utilizes multiple CNN streams to process different information from videos in parallel, such as RGB images, optical flow, and pose data. These streams are then combined to make a final prediction about the sign. Camgoz et al. [10] introduced a Transformer model for sign language recognition and translation. The model uses an encoder–decoder architecture, where the encoder processes the input video, and the decoder generates the corresponding sign sequence or translation sentence. Zhao et al. [11] leverages BERT's pre-training success for sign language recognition (SLR) by modeling hand and body poses as triplet units within a Transformer architecture. The recent works by Vu et al. [7, 12, 13] introduce multi-modal models that integrated information from various sources, demonstrating a significant improvement in model accuracy resulting from combining these diverse information sources.

ViViT (Video Vision Transformer) [14] is a state-of-the-art deep learning model designed for video understanding tasks. It extends the Vision Transformer (ViT) architecture to process spatiotemporal data by modeling video frames as sequences of image patches. The authors of ViVit have found that using smaller input tubelet sizes, which results in more tokens and finer details, consistently improves performance across all model architectures. However, this comes at the cost of increased computational complexity measured in floating-point operations (FLOPs). Another motivation behind study is the fact that ViVit also utilizes the ability to find correlations among all tokens through Query, Value and Key system of Vision Transformer forming a shared information system for tokens between multiple branches. Our study presents CrossViViT, a novel multi-stream model based on the Video Vision Transformer (ViViT). The prohibitive computational costs associated with fine-grained token processing in ViViT-H and the insufficient feature capture for fine-grained activity recognition in ViViT-S motivated us to design a two-branch architecture that can overcome the information loss typically

encountered in pyramid structures such as SlowFast networks [15] during the training process.

This paper has the following primary contributions:

- Development of a dual-branch ViViT architecture that incorporates fine-grained and coarse processing streams, enabling comprehensive temporal and spatial feature extraction while preserving spatial information integrity.
- Introduction of an optimized training methodology incorporating auxiliary loss parameter adjustment, specifically designed to enhance model accuracy for medium-scale sign language datasets.
- Comprehensive empirical validation through extensive experiments across multiple distinct datasets provides a thorough analysis of the effectiveness of the transformer-based multiview approach.

The remainder of this paper is structured as follows: Section 2 presents a comprehensive review of related literature, Sect. 3 details the proposed model architecture, Sect. 4 describes the experimental methodology and results, and Sect. 5 concludes with a discussion of findings and future research directions.

2 Related works

2.1 Sensor-based sign language recognition

Sensor-based sign language recognition (SSLR) usually involves using specialized equipment, such as data gloves, smart watches, or motion capture systems, to capture the precise movements and gestures of sign language. Sensor-based methods can achieve high recognition accuracy, but they are often limited to specific, controlled environments and require specialized hardware. SSLR involves using sensors such as accelerometers [16], gyroscopes [17], gloves [18, 19], and flex sensors [20] to capture the movement and orientation of the hands and fingers during sign language gestures. The data captured by the sensors is then processed and analyzed using machine learning algorithms to recognize the sign language gestures. SSLR has shown promising results in improving the accuracy and robustness of SLR systems. Wu et al. [21] developed an ultra-sensitive and robust strain sensor that can accurately recognize sign language gestures with high accuracy. They used the principal component analysis and traditional machine learning models to classify the sign language gestures. Pezzuoli et al. [22] also employed data gloves to capture the hand movements to translate into sign language words. The authors also comprehensively compared machine learning algorithms to classify sign language gestures. Random forest (RF) algorithm was implemented in Byberi et al. [23] to recognize American sign language gestures, achieving high accuracy and real-time performance. Dynamic time warping has also been used in the SSLR system to match the temporal variation of sign language gestures [24]. They used a combination of hand movement and hand shape to recognize sign language words. Amin et al. [25] trained an Artificial neural network (ANN) to recognize isolated static hand postures of American sign language. Their training dataset consisted of hand posture images captured using a customized data glove. Besides mentioned traditional machine learning models, deep learning models have been explored to recognize sign language words from sensor data. Shi et al. [26] proposed a 1DCNN model to recognize gesture data from magnetic sensors. Delpreto et al. [27] incorporated the LSTM model to extract temporal features from flexible

sensors and accelerometers to recognize sign language words. Their proposed model can achieve an accuracy of 96.3%. Gu et al. [28] combined CNN and LSTM to enable end-to-end translation from sensor signals to text sentences. LSTM and dense network have been integrated in Lee et al. [29] to design a gesture classification network. Their approach could recognize gestures and compress gesture sequences simultaneously. While showing many potential, sensor-based models for sign language recognition are constrained by various limitations. These include the need for sophisticated hardware, vulnerability to noisy data, and the requirement of controlled settings, rendering them unsuitable for real-world applications.

2.2 Vision-based sign language recognition

In contrast to sensor-based methods, vision-based sign language recognition (VSLR) aims to recognize sign language gestures using cameras and computer vision techniques. Compared to sensor-based approaches, vision-based methods offer a more natural and unobtrusive interaction and can potentially be deployed in a wider range of real-world scenarios [30]. Rather than relying on specialized equipment, VSLR systems typically leverage readily available cameras and advanced computer vision algorithms to detect, track, and recognize sign language gestures from video data. Early VSLR systems often relied on hand-crafted feature extraction techniques, such as Histogram of Oriented Gradients (HOG) and Scale-Invariant Feature Transform, combined with traditional machine learning classifiers like Support Vector Machines and Hidden Markov Models [31, 32]. Raghuveera et al. [33] applied HOG feature extraction to retrieve features from RGB image and SVM model to classify Indian sign language. SIFT feature extraction technique was used in Arooj et al. [34] to extract key points from sign language images and feed them into the SVM classifier. On the other hand, Solis et al. [35] used Jacobi-Fourier moments as visual features and SVM for classification. They achieved 95% accuracy on the Mexican sign language dataset.

Although these earlier vision-based approaches demonstrated promising results, they relied heavily on manual feature engineering and struggled to handle the complexities of unconstrained real-world sign language gestures. In recent years, rapid advancements in deep learning have revolutionized the field of vision-based sign language recognition. [36] applied a CNN model on the segmented Korean sign language dataset and achieved significant improvements compared to traditional machine learning approaches. [37] proposed a framework for Chinese sign language recognition data. The 3D convolutional residual network was used to extract spatial features of the video frame, while temporal features were captured by an LSTM model with Connectionist temporal classification (CTC) loss. The author of [38] introduced a sign language recognition framework that involves extracting spatial–temporal features using a 3D convolutional neural network and selecting highlight features using an attention model.

One of the most prominent methods based on 3D convolutions in the field of video recognition is the SlowFast model, which leverages a multiview and dual-branch architecture including an amount of 3D CNN blocks, which was designed to extract rich features in both the slow and fast branches. This approach captures high-level temporal semantics and fine-grained motion details across different temporal resolutions. A core limitation of pyramid-based models like SlowFast is that as video data passes through the multiple layers of 3D CNNs, much of the spatial information is progressively reduced.

This is due to the high depth required for rich feature extraction, which can lead to a loss of fine spatial details, potentially impacting the model's effectiveness in capturing rich spatial features necessary for video understanding.

Recently, transformer-based models have also shown promising results. Given the success of ViTs in face anti-spoofing tasks [39–41], future VSLR research could explore integrating similar architectures to improve feature extraction and classification in sign language recognition tasks. De Coster et al. [42] used a multi-head attention mechanism in their transformer model to capture long-range dependencies in the sign language video. They achieved state-of-the-art performance on the Flemish Sign language corpus with an accuracy of 74.7%. In Kothadiya et al. [43], the vision transformer model was implemented to recognize static Indian sign language. They divide the frame of sign language video into a series of positional embedding patches, which are forwarded to a transformer block with four self-attention layers. Pu et al. [44] proposed a feature-isolated mechanism to capture local spatial–temporal context. This approach can model context information independently and simultaneously from individual features, thus improving the robustness of SLR models. Transformer models not only enable direct sign language recognition but also facilitate the detection of word boundaries in sign language sentences, acting as a foundational component for sentence-level recognition [45, 46]. Despite achieving state-of-the-art results in sign language recognition, transformer models are computationally expensive due to their large number of parameters and the need for extensive data and training time. This has limited their practical applications. To mitigate this issue, we propose CrossViViT, a model designed to reduce computational costs while preserving performance by dividing the processing pipeline into multiple branches and combining their final results to classify the sign language videos.

Theoretically, such a hierarchical structure is not essential for transformers, as each token inherently"attends" to all other positions. However, in practice, multiscale processing techniques have been applied to transformers to reduce model complexity and compensate for limited training data, proving to be effective. Our model deviates from the traditional pyramid structure by employing a multiview approach: it directly ingests different views of a video and processes them through cross-view encoders. Experimental results consistently show that this multiview architecture outperforms single-view models regarding accuracy and efficiency (as measured by FLOP trade-offs). This superior performance arises because processing multiple views in parallel provides more considerable accuracy improvements than increasing the transformer's depth alone. Notably, these accuracy improvements persist even as we scale the model's capacity to over a billion parameters, a result not previously demonstrated by pyramid-structured transformers. While prior works [47–49] operate in domains outside sign language recognition, their use of lightweight Transformer-CNN hybrids and attention-enhanced architectures provide insights into effective multi-modal and ensemble-based learning. Our proposed Cross-ViViT draws upon similar principles-namely multi-branch processing and hierarchical attention, but is uniquely adapted to handle spatial–temporal video data from sign language gestures, emphasizing fine-grained motion features across multiple views.

3 Proposed method

In practice, rich information in video data often appears at multiple levels of detail and scale, requiring models to capture features across these levels. However, multiview models can become computationally expensive if they process data across multiple scales simultaneously. We propose a cost-efficient multiview model that maintains accuracy while reducing computational demands to address this. Our approach involves splitting the model into two branches: a coarse (primary) branch and a fine-grained (supplementary) branch. The coarse branch, designed to extract features with minimal computational cost, incorporates multiple Transformer layers to improve recognition over the entire dataset, and the fine-grained branch, which is more computationally intensive, includes only a few Transformer layers. This structure allows the coarse branch to capture broad data insights while the fine-grained branch selectively enhances details at specific levels without substantially increasing computational costs. After all, we use an exchange information method to merge rich information from two branches and take it into a FCN that returns one-hot vector. In here, we use Cross Entropy Loss twice, the loss in latest block can be calculated to back-propagate through this stream and the other is considered as an auxilary loss which prevent backbone I3D from vanishing gradients problem while training the model. An example of this overall procedure is illustrated in Fig. 1

Before analyzing the details of our architecture, we begin with an overview of a transformer base model, which is the main backbone of our method.

3.1 Preliminaries: vision transformer

Let $V \in \mathbb{R}^{T\times H\times W\times C}$ represent the input video. Transformer-based architectures [45] process these inputs by transforming them into discrete tokens, which are subsequently passed through multiple transformer layers sequentially. Vision Transformer (ViT) [50] partitions an image into non-overlapping patches, projecting each linearly to create tokens. ViViT generalizes this approach for videos by segmenting the video into N non-overlapping spatiotemporal"tubes"from the input, defined as $x_1, x_2, \ldots, x_N \in \mathbb{R}^{t\times h\times w\times c}$, where

$$N = \left\lfloor \frac{T}{t} \right\rfloor \times \left\lfloor \frac{H}{h} \right\rfloor \times \left\lfloor \frac{W}{w} \right\rfloor .$$

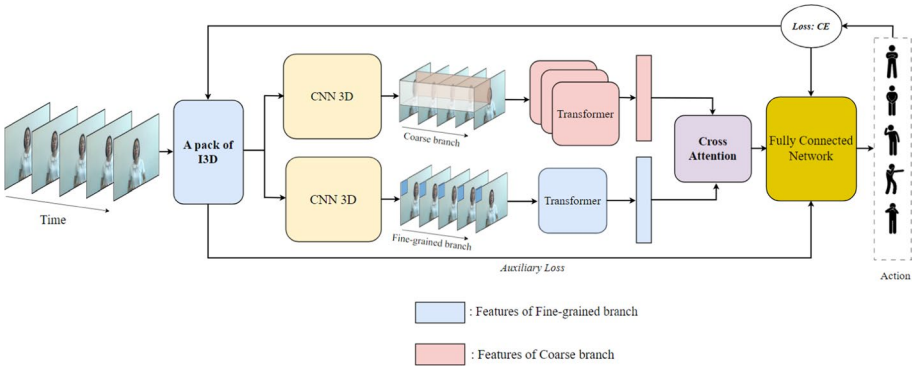


Fig. 1 Flowchart of our proposed approach

Each tube x_i is projected into a token $z_i \in \mathbb{R}^d$ using a linear operator E, such that $z_i = Ex_i$. These tokens are concatenated into a sequence, prepended with a learnable class token $z_{\text{cls}} \in \mathbb{R}^d$ [51]. Since transformers are permutation-invariant, a positional embedding $p \in \mathbb{R}^{(N+1)\times d}$ is added to the sequence. The tokenization process is represented as:

$$z^0 = [z_{\text{cls}}, Ex_1, Ex_2, \ldots, Ex_N] + p.$$

Notably, the linear projection E can also be considered a 3D convolution with a kernel size of $t \times h \times w$ and a stride of (t, h, w) in the temporal, height, and width dimensions, respectively.

The token sequence z is then processed by a transformer encoder with L layers, where each layer ℓ sequentially applies the following operations: $y^{(\ell)} = \text{MSA}(y^{(\ell-1)})$.

3.2 Proposed detail model

Based on Vision Transformer's theoretical framework, utilizing a token dimensionality of *14*14*4* yields superior accuracy metrics compared to a *16*16*4* token configuration. However, when scaling to *32*32*4* token dimensions, the computational complexity, as measured in floating-point operations per second (FLOPS), increases by a factor of three, indicating a significant computational overhead [14]. Larger patches offer computational efficiency but may miss subtle visual cues, while smaller patches are more sensitive to fine-grained variations at the cost of higher complexity. Our hybrid strategy leverages the strengths of both scales-promoting accuracy and speed-by enabling bidirectional information flow between the two branches via a cross-attention mechanism.

3.2.1 Patch embedding

The architecture implements a dual-branch topology, wherein each branch comprises a distinct configuration of Transformer layers, optimized for their respective token input dimensions. Figure 2 provides a detailed architectural schematic and delineates the optimal hyperparameters for maximizing the efficacy of the dual-branch configuration. The architecture consists of two primary branches: (1) the Coarse Branch and (2) the Fine-Grained Branch. Each branch employs a specialized Patch Embedding mechanism to decompose the input frame into discrete tokenized representations. The Coarse Branch is engineered to operate with an expanded tube size receptive field (e.g., *4*16*16*256*), while the fine branch is optimized for granular feature extraction utilizing a reduced tube

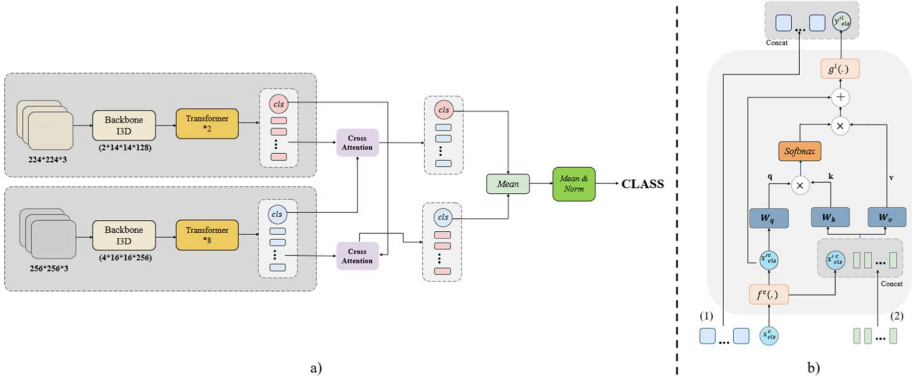


Fig. 2 **a** Overview of our proposed architecture, **b** details of cross attention module

size configuration (e.g., *2*14*14*128*). Our patch partitioning strategy is also inspired by the concept of multi-view perception commonly found in object detection models such as YOLO [52]. In such models, small-scale features are often only detectable from finer-grained views. Drawing from this idea, we interpret the use of different patch sizes in our dual-branch Transformer as a form of multi-view modeling.

Empirically, we observed that using significantly smaller patches (*e.g.,* $4 \times 4 \times 2$) does not produce meaningful improvements in accuracy while introducing a considerable increase in computational cost. However, excessively large patches (*e.g.,* $32 \times 32 \times 4$) lead to severe information loss, which compromises the accuracy of the model. Through extensive validation, we found that $16 \times 16 \times 4$ and $14 \times 14 \times 2$ for the two branches are the most optimized configuration. This optimized configuration allows the coarse branch to retain the global semantic structure with low computational overhead, while the fine branch contributes essential local detail. This configuration achieves an effective trade-off between accuracy and efficiency, making it particularly suitable for real-time sign language recognition tasks.

$$\begin{aligned} &x' = F_{I3D}(x), \quad x \in \mathbb{R}^{H \times W \times T}, \quad x' \in \mathbb{R}^{\frac{H}{h} \times \frac{W}{w} \times \frac{T}{t}} \\ &where: N_t = \frac{T}{t}, \quad N_h = \frac{H}{h}, \quad N_w = \frac{W}{w} \end{aligned} \tag{1}$$

Architectural considerations become paramount given the relatively modest scale of the Sign Language Recognition (SLR) dataset. In Eq. 1, we implement a modified backbone architecture inspired by the I3D framework as our patch embedding mechanism, superseding conventional linear block implementations. This design choice is motivated by empirical evidence suggesting that large-scale Transformer architectures encounter convergence difficulties when trained on medium-scale datasets, primarily due to their limited capacity to learn local feature representations during initial training phases. Consequently, we opted for a CNN-based backbone utilizing the I3D architecture, which augments our proposed model's capacity for local feature extraction. Given the permutation-invariant nature of transformer architectures, we incorporate a positional embedding $\mathrm{p} \in \mathbb{R}^{(N+1) \times d}$ into the sequence. The tokenization process can be formally expressed as:

$$z^0 = [z_{\text{cls}}, x'_1, x'_2, \ldots, x'_N] + p. \tag{2}$$

3.2.2 Multi-view attention

Following the tokenization of both branches with distinct tube size configurations, we employ multiple Transformer layers for feature extraction, as depicted in Fig. 2. To optimize the trade-off between the accuracy and computational complexity, we implement θ layers ℓ in the coarse branch, as formalized in Eq. 3.

$$y_c^\ell = \mathrm{MSA}(\mathrm{LN}(z_c^\ell)) + z_c^\ell, \quad z^{\ell+1} = \mathrm{MLP}(\mathrm{LN}(y_c^\ell)) + y_c^\ell \tag{3}$$

with y_c^ℓ as the coarse branch, $\ell \in [0, \theta]$
and $\theta \cdot \frac{1}{\lambda}$(e.g., $\lambda = 4$) for fine-grained layers (ℓ') illustrated in 4.

$$y_f^{\ell'} = \mathrm{MSA}(\mathrm{LN}(z_f^{\ell'})) + z_f^{\ell'}, \quad z^{\ell'+1} = \mathrm{MLP}(\mathrm{LN}(y_f^{\ell'})) + y_f^{\ell'} \tag{4}$$

with $y_f^{\ell'}$ as the fine-grained branch, $\ell' \in \left[0, \frac{\theta}{\lambda}\right]$.

Equations 3 and 4 compare and contrast the imbalance of layers between coarse and fine grain. As previously established, the model's performance is predominantly governed by the computationally efficient coarse branch, which incorporates a substantial number of attention layers. This architectural decision significantly enhances the feature extraction capabilities of the coarse branch, enabling it to maintain robust accuracy metrics even in autonomous operations. Conversely, the fine-grained branch serves primarily as a supplementary mechanism for high-resolution feature extraction rather than functioning as an independently operational component within the overall architectural framework.

3.3 Cross view attention

The architecture implements a fusion class token methodology to facilitate multiview information exchange between pathways through cross-attention, as illustrated in the right portion of Fig. 2. This cross-attention mechanism introduces an innovative approach within the Transformer Decoder framework, enabling inter-branch feature extraction through cross-branch querying operations.

The class tokens $z^{[\mathrm{cls}]}$ and $z^{[\mathrm{patch}]}$, generated from the terminal transformer layers of both branches, serve distinct purposes. The class tokens $z^{[\mathrm{cls}]}$ are theoretically demonstrated to encapsulate comprehensive global features of their respective branches. In the cross-branch information exchange process, the coarse-grained branch's class token functions as a Query $Q = z_c^{[cls]} W_q$, seeking global feature information from the fine-grained branch, which provides the Key $K = z_f W_k$ and Value $V = z_f W_v$ components.

$$\begin{aligned} A &= \mathrm{softmax}\left(\frac{qk^T}{\sqrt{C/h}}\right), \\ CA(z) &= Av, \end{aligned} \tag{5}$$

where C and h are the embedding dimension and the number of heads.

Equation 5 delineates the utilization of cross-branch class tokens as attention queries, termed Cross Attention (CA). Given that contemporary transformer architectures typically employ multi-head attention mechanisms (MCA) for feature extraction, the comprehensive formulation is expressed as:

$$\begin{aligned} y_{fu}^{[cls]} &= f^l\left(z_c^{[cls]}\right) + MCA(LN([f^l(z_c^{[cls]})||z_f^{[patch]}]) \\ z_c^l &= [g^l(y_{fu}^{[cls]})||y_c^{[patch]}] \end{aligned} \tag{6}$$

with $f^l(.)$ and $g^l(.)$ are the projection for dimension alignment

Equations 5 and 6 demonstrate the fusion methodology utilizing class tokens as cross-branch queries, enabling efficient information aggregation from the fine-grained branch to the coarse branch while maintaining computational efficiency. In this formulation, f^l and g^l represent linear transformations that facilitate appropriate dimensional alignment of class tokens between branches.

3.4 Auxiliary loss

Our analysis revealed that integrating an I3D-based backbone with Transformer networks occasionally results in information degradation during the bidirectional propagation due to the vanishing gradient problem. To mitigate this phenomenon, we introduce

an auxiliary loss function at the backbone's terminal layer, ensuring consistent information preservation across the network during the training phase. The auxiliary loss function is formulated as:

$$\mathcal{L}_{aux} = \lambda \cdot \mathcal{L}_{task}, \tag{7}$$

where the primary task loss, $\mathcal{L}_{task}$, is formulated as:

$$\mathcal{L}_{\text{task}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \log \hat{y}_{i,c}.$$

The constituent parameters are defined as follows:

- N is the number of samples in the batch,
- C is the number of classes,
- $y_{i,c}$ represents a binary indicator function, equal to unity when sample i belongs to class c, and zero if otherwise.
- $\hat{y}_{i,c}$ represents the model's predicted probability distribution over class c for sample i, derived from the soft-max activation.

Here, $\mathcal{L}_{aux}$ represents the auxiliary loss, $\mathcal{L}_{task}$ is the primary task loss, and λ is a scaling factor that controls the contribution of the auxiliary loss to the overall objective. This formulation ensures that the information flow remains consistent throughout the backbone and facilitates more effective gradient propagation during training.

4 Experiments

4.1 Dataset detail

We conduct our experiments on three public datasets: Vietnamese SignLanguage (SLR) [1], Something-Something v2 (Sth-v2) [53] and a large-scale dataset for Word-Level American Sign Language (WLASL) [54]. While VSL is a Vietnamese sign language dataset that includes gestures representing words in sign language, the Something something v2 is a dataset collected from fine-grained actions to build and evaluate video understanding and classification models.

4.1.1 VSL

The Vietnamese Sign Language (VSL) dataset is captured using a camera. This dataset advances research in HAR, supports sign language standardization, and promotes its educational use across countries. The VSL dataset was specifically designed with deep learning methods, considering factors such as data variety, real-world environments, and standard equipment usage. Recognizing the significant impact of medical communication for the deaf-mute community, particularly in emergency healthcare settings, the dataset includes sign language words related to healthcare and clinical topics. The importance of such datasets has been further emphasized by the COVID-19 pandemic, which highlighted the critical need for automatic sign language recognition (SLR) systems to facilitate communication in medical contexts.

Although the current dataset is in its initial phase, it lays the foundations for collecting a more comprehensive dataset. Future expansions will include additional domains, broadening the dataset's applicability. The VSL dataset is also designed to be scalable,

with the potential to expand to more than 10,000 classes, further enhancing its utility for research and practical applications in sign language recognition. On average, each label has 130–150 videos. There are a total of 5 different contexts and performers, with each context accounting for about 20–30 videos. To prevent the model from overfitting to background cues, we select an average of 50 videos per class that share a similar background and ensure these specific backgrounds do not appear in the training set which enforces the model to rely on the signer's motion and features rather than spurious correlations from the background, leading to a more robust and generalizable sign language recognition system.

4.1.2 Sth-v2

The Sth-v2 is an extensive collection comprising 220,847 labeled video clips of 174 classes. Each clip captures a human performing one of several pre-defined, basic actions with everyday objects. These actions encompass fine-grained and nuanced hand gestures, such as"putting something into something","turning something upside down", and"covering something with something". The dataset is meticulously designed to facilitate training of machine learning models for improved understanding and recognition of subtle human-object interactions and hand gestures. To evaluate our proposed model's generalizability to large-scale action recognition tasks, we included the Sth-v2 dataset in our experiments. This dataset is known for its fine-grained annotations and extensive class diversity. Given that our model is designed for sign language recognition, which often involves a much larger class space, we believe this dataset provides a rigorous test of our model's scalability.

4.1.3 WLASL

WLASL (Word-Level American Sign Language) is the largest publicly available dataset for word-level ASL recognition, designed to facilitate research in sign language understanding using deep learning and computer vision techniques. WLASL consists of over 21,000 video clips covering more than 2000 distinct ASL words. These signs were collected from real-world video sources, therefore, WLASL provides a challenging yet realistic benchmark for automatic sign language recognition, offering variations in lighting, background, and signer styles. The dataset is structured into subsets of different vocabulary sizes, such as WLASL100, WLASL300, WLASL1000, and WLASL2000, making it suitable for various levels of model training and evaluation. Details about all of the datasets are presented in Table 1

4.2 Experiements setup

Our proposed methodology and all comparative architectures underwent empirical evaluation across multiple datasets: the previously discussed VSL dataset and the widely

Table 1 Statistics for the datasets

Dataset	Class numbers	Video length (s)	Number of videos
VSL	50	8	9000
Sth-v2	174	2–6	220,847
WLASL-100	100	1–4	2038
WLASL-300	300	1–4	5118
WLASL-2000	2000	1–4	21,000

Table 2 Preprocessing Settings

Preprocessing	Train	Validation
Center crop	224×224	224×224
Mix up	0.2	No
Resize import	256×256	256×256
Normalize	Yes	Yes
Color jilter	Yes	No

Table 3 Hyperparameter setup

Parameters	Value
Learning rate	5e−6
Learning rate decay	0.1
Weight decay	5e−2
Epoch warm up	10
Momentum	0.999
Smoothing	0.1
Depth	6
Drop out	0.5
Drop path	0.2
Norm batch	1e−6
Epoch	50
Batch size	8
Patch size	[14, 16]
Tube size	[2, 4]
Embed_dim	[128, 256]
Number of labels	50

recognized fine-grained action recognition benchmark, Smth-v2. The datasets were partitioned into training and validation subsets with ratios of 8:2 and 7:3, respectively. Before model training, comprehensive data preprocessing protocols were implemented to ensure data quality and consistency which is shown on Table 2.

The experimental framework was implemented utilizing PyTorch Framework 2.1.0 and Python 3.10. All computational experiments were conducted on hardware infrastructure consisting of an NVIDIA M10 GPU with 24GB VRAM capacity and an Intel Xeon processor operating at 2.1 GHz. The training phase was executed with the following hyperparameters and optimization strategies in the Table 3:

4.3 Results

In this section, we design our empirical study to answer key research questions that guide the evaluation of our proposed method. These questions serve as a foundation for systematically analyzing our model's performance and understanding its strengths and limitations.

Q1: How does our approach perform on the sign language recognition task?
Q2: How difficult is it to recognize each word in the VSL dataset?
Q3: What is the impact of backbones on overall performance?
Q4: How do branch configurations and fusion strategies affect recognition performance?
Q5: How does our model perform on the large scale sign language dataset?
Q6: What challenges does our model need to address?

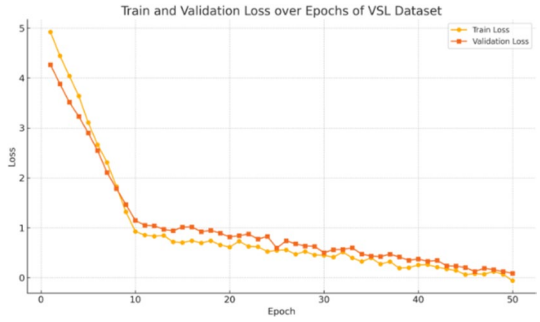


Fig. 3 Training and validation loss over epochs for VSL dataset

Table 4 Comparison between similar models

Model	Backbone	Acc@1 (%)	Precision	Recall	F1 score (%)	FLOPS
ViVit_B	*None*	85.7	86.3	82.5	84.4	455.2
ViVit_L	*None*	95.8	95.2	94.6	94.9	3992
Mae	*Vit_B*	89.9	89.1	87.5	88.3	–
I3D	*None*	74.58	71.3	72.9	72.1	312.2
CrossViVit	***I3D***	***92.47***	***90.9***	***92.3***	***91.6***	***629.35***
CrossViVit-auxiliary	***I3D***	***95.7***	***93.3***	***96.1***	***94.7***	***629.35***

Q7: How does our model perform on other fine-grained datasets?

(Q1): How does our approach perform on the sign language recognition task?

A fundamental aspect of evaluating any machine learning model is its performance compared to existing methods. To demonstrate the effectiveness of the model in addressing the sign language recognition task, we conducted experiments on a medium-sized dataset, namely VSL, which contains 50 labels encompassing everyday actions such as"congratulate," "clap,"and others.

Additionally, as shown in the loss curve in Fig. 3, both training and validation loss decrease consistently over epochs with minimal gap, indicating that the model generalizes well and does not suffer from overfitting during training.

The evaluation of different models includes metrics such as accuracy and Floating Point Operations (FLOPs) for each model on the VSL dataset.

The results in Table 4 clearly demonstrate that the **CrossViVit** model outperforms other models, such as ViVit-B, I3D, and Mae, achieving an accuracy of approximately 92.47% and Precision is 90.9%, Recall is 92.3%, F1 score is 91.6%. Notably, incorporating Auxiliary Loss (**CrossViVit-Auxiliary**) during training further boosts the model's accuracy to 95.7%, which is only slightly lower than that of the ViVit-L model (95.8%). However, compared to ViVit-L, our model presents a significant advantage in computational efficiency, requiring only 629 million FLOPS—substantially lower than that of ViVit-L.

The difference in FLOPS primarily arises from the model's design, wherein the majority of processing is concentrated in the coarsed branch with an input tube of *16*16*4*, while only partial information is utilized from the fine-grained branch with an input of *14*14*2*. This design significantly reduces the computational load, making the model much lighter than ViVit-L and H models, which require full processing of a *14*14*2* input tube.

Here, we observe that while the overall performance of the model on the VSL dataset is highly satisfactory (95.3%), a closer examination of individual word evaluations reveals inconsistencies in recognition accuracy across different words. This suggests that performance at a broader scale does not necessarily translate to uniform success across all sign gestures, highlighting the need for a more granular analysis. That is exactly what we will do in our next question.

(Q2): How difficult is it to recognize each word in the VSL dataset?

This finding suggests that there are underlying inconsistencies in how the model understands and distinguishes certain words, indicating potential issues in the learning process. Sign language application poses significant challenges due to the subtle variations in movements, where even slight deviations can impact interpretation. In the subsequent analysis, we will conduct an F1 Score evaluation for each label of the SLR dataset to systematically examine the specific classification challenges encountered by the model for each sign language gesture in the dataset. This label-wise assessment will help identify areas where the model may struggle with precision or recall, providing insights for targeted model improvement.

Table 5 shows that the classification issues are particularly concentrated in certain labels such as "Lost," "Close," "Teach," and "Declare." A closer inspection of the confusion matrix (Fig. 4) reveals frequent misclassifications among these labels, contributing to their significantly lower F1 scores, typically hovering around 0.7%. Upon further analysis, it becomes apparent that these labels share similar characteristics, particularly in their actions, which closely resemble one another. For instance, the "Lost" and "Close" labels only differ in the reversal of the final segment of the gesture. These similarities in action also challenge the model in distinguishing between these labels.

Therefore, we had to conduct a more in-depth analysis of the model's learning process to better understand how it differentiates between specific words in the next research question.

(Q3): What is the impact of backbones on overall performance?

The requirement for large datasets to achieve convergence and optimal performance in Transformer-based architectures presents a significant challenge when training on medium-scale datasets like VSL. We propose a custom backbone architecture inspired by previous state-of-the-art models to address this issue. Table 6 summarizes the experimental results of applying different backbone architectures to our model.

Table 6 compares and contrasts the result of each backbone on SLR datasets. Lightweight backbones such as R2Plus, CNN3D, and I3D were chosen rather than larger architectures. This choice aligns with our objective of ensuring fast inference speed for real-world sign language recognition tasks. These results prove that I3D outperforms other backbones at 92.47%. While we have also experimented with more extensive backbones, such as SlowFast and ViT-S, we found that although they provided slight improvements in accuracy, they incurred significantly higher computational costs, which contradicts our initial goal of balancing computational efficiency and model performance.

Additionally, we conducted experiments with optimization techniques and loss adjustments, including the EMA (Exponential Moving Average) model and the

Table 5 F1 score evaluation for each label

Index	Word	F1 score
1	Celebrate	0.9841
2	Appease	0.9206
3	Apply	0.9643
4	wonder	1.0000
5	Close	0.8438
6	Patient	0.7308
7	Know	0.9787
8	Present	1.0000
9	Ministry of health	0.9737
10	Isolation	0.9143
11	Temptation	1.0000
12	Thank	0.8657
13	Need	0.8085
14	Accept	0.8710
15	Run	0.9615
16	Wait	1.0000
17	Pain	0.9231
18	Teach	0.7895
19	Hate	0.9333
20	Help	0.9123
21	Fan	0.8219
22	Sacrifice	0.9153
23	Cough	0.9841
24	Get married	0.8302
25	Declare	0.9118
26	Face mask	0.9412
27	Isolation area	0.9508
28	Infect	0.9565
29	Worried	0.9655
30	Rest	1.0000
31	Mind	0.9697
32	Backbite	0.8627
33	Vomiting	0.9538
34	Punish	0.7778
35	Interview	0.9474
36	Recover	0.9375
37	Share	0.7869
38	Fever	0.9677
39	Exercise	0.8788
40	Visit	0.9123
41	Lost	0.5200
42	Like	0.9787
43	Affectionate	0.9286
44	Incubation	0.9552
45	Donate	0.8378
46	Obey	0.9863
47	Virus	1.0000
48	Ask for permission	1.0000
49	Discharge from hospital	0.9730
50	Love	1.0000

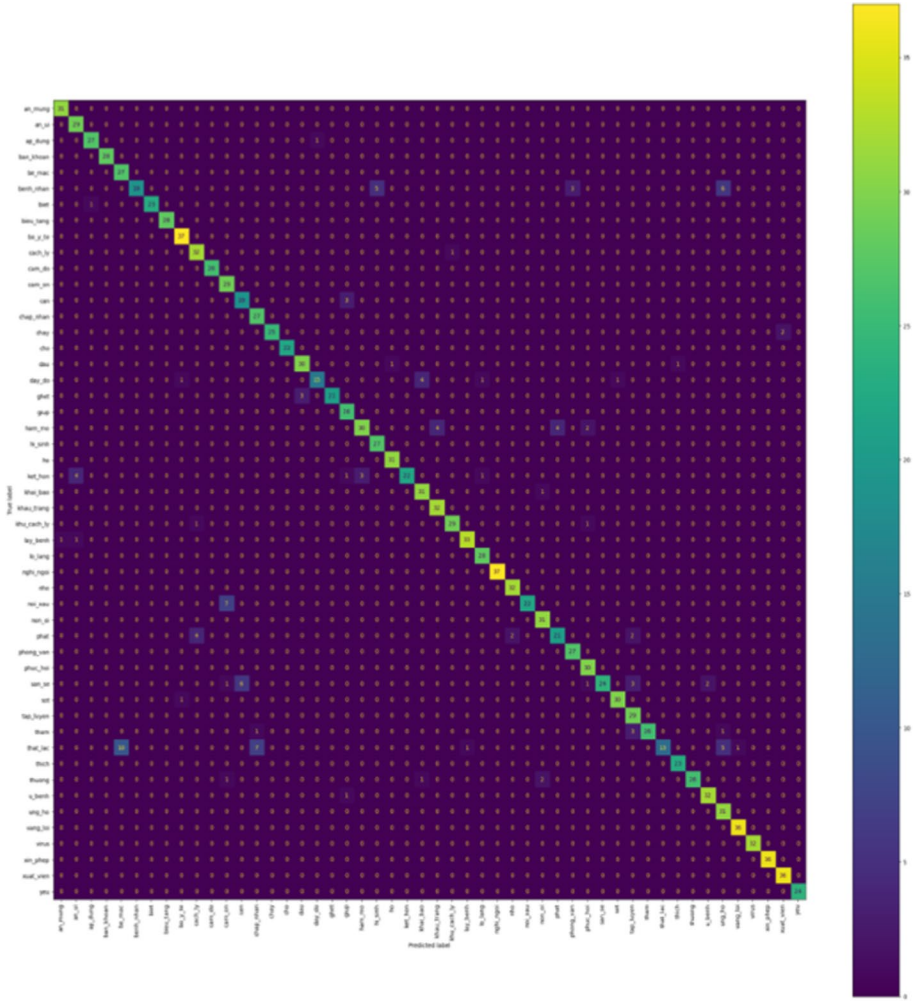


Fig. 4 Confusion matrix for all labels

Table 6 Comparison of various backbones

Backbone	Acc@1 (%)
I3D	92.47
R2+1	87.6
R3D	89.2
CNN3D	63.6

Table 7 Comparison on various optimization methods

Optimization methods	Acc@1 (%)
Auxilary-loss	95.7
EMA	92.57
SmoothingCrossEntropy	92.47

SmoothingCrossEntropy loss function, to enhance the model's performance. Table 7 presents the results obtained after applying these techniques.

The Fig. 5 demonstrates that, in the absence of auxiliary loss, the backbone undergoes minimal updates, primarily relying on the pre-trained knowledge from Kinetics-400. In contrast, applying an independent auxiliary loss to the I3D backbone forces

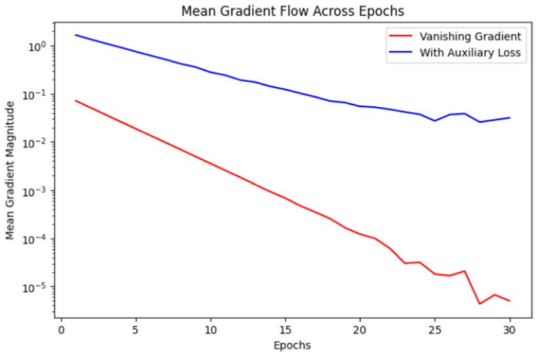


Fig. 5 Mean gradient flow with and without auxiliary loss

Table 8 Evaluation on each branch

Model	Acc@1 (%)
Large branch	91
Small branch	80
Fusion branches	92.47
Fusion branches-auxiliary	95.7

Table 9 Effect of varying Transformer layer counts on SLR accuracy (VSL dataset)

Configuration (fine, coarse, cross-attn)	Accuracy (%)
CrossVivit (FineGrain-1 and Coarse-6 and CrossAttention-1)—recommended	92.47
CrossVivit (FineGrain-1 and Coarse-8)	92.53
CrossVivit (FineGrain-1 and Coarse-4)	90.40
CrossVivit (FineGrain-2 and Coarse-6)	91.80
CrossViVit (CrossAttention-2)	92.10

it to optimize its feature extraction specifically for the VSL dataset. This results in a 3% improvement in accuracy compared to the baseline model without auxiliary loss.

(Q4): How do branch configurations and fusion strategies affect recognition performance?

To illustrate the effectiveness of integrating and exchanging information between the two branches based on Cross-view Attention, we conducted experiments that involved separating the CrossViVit into two independent models, resulting in each branch functioning as a standalone model, which is utilized based on Self Attention Mechanism. The table presented below delineates the evaluation results for each branch individually.

Our approach prioritizes the classification of sign language gestures utilizing the coarse branch, facilitating significant reductions in computational costs. In contrast, the fine-grained branch is designed to capture more nuanced information that the coarse branch may overlook, thus enhancing the model's overall performance. Table 8 indicates that when the fine-grained branch operates independently, its accuracy reaches approximately 80%, whereas the coarse branch achieves an accuracy of up to 91%. Furthermore, when information from both branches is integrated, the model's accuracy improves by 1.47% compared to the coarse branch alone. This finding underscores the ability of the fine-grained branch to identify features that the coarse branch may not detect (Table 9).

To address concerns regarding architectural rigidity, we conducted a series of ablation studies by varying the depth of each branch and adjusting the fusion mechanism.

Table 10 Training time and Inference speed

	GPU	CPU
Training time	0.11 s per video (RGB only)	1.52 s per video
Inference speed	0.038 s per video (RGB only)	0.43 s per video

Table 11 Performance comparison with other Transformer model on WLASL dataset

Model	Modalities		WLASL accuracy (%)		
	RGB	Skeleton	WLASL100	WLASL300	WLASL2000
SignBert (2023) [55]	✓	✓	85.05	–	–
Siformer (2024) [56]	✓	✓	86.5	–	–
SignBert (2022)	✓	✓	83.3	75.27	54.69
I3D	✓		67.01	56.24	32.48
TCK	✓		77.55	68.75	48.49
CrossViVit	✓		86.42	75.94	55.1

Specifically, we tested changes in the number of Transformer layers in the Coarse, Fine-Grained, and Cross-Attention branches. Results are summarized below:

The results show that the Coarse branch-responsible for learning high-level spatiotemporal patterns-is the most sensitive to depth changes. Increasing it from 6 to 8 layers gives only a marginal gain, whereas reducing it to 4 layers significantly drops performance, indicating underfitting. Meanwhile, increasing depth in the Fine-Grained or Cross-Attention branches has a smaller effect, validating our design that treats them as complementary components rather than primary feature extractors.

Regarding fusion design, we use a Cross-Attention mechanism to enable contextual feature exchange between branches. Although this strategy is effective, we acknowledge that we have not yet explored other fusion alternatives such as concatenation, additive fusion, or gating. These are promising directions for future work to explore more adaptive and potentially lightweight integration schemes.

(Q5):How does our model perform on the large scale sign language dataset

To evaluate the effectiveness of our proposed model's generalization ability, we conduct experiments and compare its performance against state-of-the-art (SOTA) models on the WLASL dataset. These evaluations were performed on a 13th Gen Intel®Core™i7-13700K processor with 12 cores, while the GPU used for assessment was an RTX4090 with 24GB of VRAM. We consider three levels of complexity: WLASL-100, WLASL-300, and WLASL-2000, corresponding to 100, 300, and 2000 labels, respectively, with a total of 21,000 videos. This evaluation allows us to assess the model's ability to understand sign languages beyond Vietnamese Sign Language. The following Table 10 shows the training time and inference speed for each videos for our model, note that the average training time also include calculating loss and backpropagation for each video.

Firstly, we will compare CrossVivit with state-of-the-art (SOTA) methods that are based on the Transformer architecture.

Table 11 shows that our model achieved high accuracy across all three evaluation levels, outperforming other Transformer-based models on WLASL-300 and WLASL-2000 with 75.94% and 55.1%, respectively. It is only slightly behind Siformer (85.6%) by a small margin (86.42%) on WLASL-100. Then, to demonstrate the model's improvement in computaional cost, we will go into detail of the speed and FLOPS with Siformer, the fastest model from the mentioned models above. Our model still maintains the same

Table 12 Detailed computational cost comparison between our method and siformer

Model	FLOPS (M)	Inference speed (s)	Acc (top 1) (%)
Siformer	622	0.0332	86.5
CrossVivit	629	0.038	86.42

Table 13 Performance comparison with other hybrid models on WLASL dataset

Model	Modalities			WLASL accuracy (%)	
	RGB	Skeleton	Flow	WLASL100 (top 10)	WLASL300 (top 10)
Fusion-3 [57]	✓	✓		90.16	86.22
GCN-BERT [58]		✓		88.67	80.93
GLR [59]	✓			96.90	84.91
hDNN-SLR [60]	✓	✓	✓	98.75	98.02
CrossViVit	✓			97.6	96.53

Table 14 Comparison with state-of-the-art cross-modal and multimodal frameworks

Model	Modalities			Dataset	Acc (top1) (%)
	RGB	Skeleton	Flow		
STSLR	✓	✓		SLR	89.4
CrossViVit (ours)	✓				**91.6**
StepNet	✓		✓	WLASL100	78.77
CrossViVit (ours)	✓				**86.42**
StepNet	✓		✓	WLASL300	75.32
CrossViVit (ours)	✓				**75.94**

inference speed and computational efficiency as Siformer, as shown in Table 12. This is achieved while only utilizing RGB information, without requiring additional skeleton and keypoint features as Siformer does.

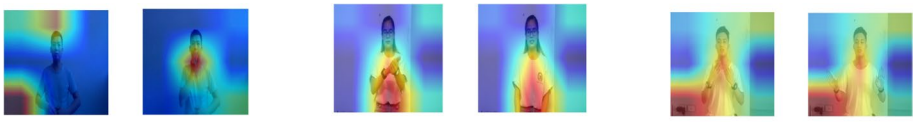
Secondly, We compared our method with hybrid models and multi-branch architectures to provide a broader perspective on the effectiveness of using multi-branch Transformer-based models. This evaluation helps assess how our approach balances accuracy and computational efficiency compared to other state-of-the-art methods.

As shown in Table 13, our proposed model achieved high top-10 accuracy on WLASL-100 and WLASL-300 (97.6% and 96.53%), slightly lower than hDNN-SLR (98.75% and 98.02%). However, unlike hDNN-SLR, our model does not rely on Flow and Skeleton data as additional input streams. This design choice makes our model more scalable and practical for real-world deployment while maintaining competitive performance.

We conducted further comparisons against cross-modal and multimodal self-supervised approaches. Recent frameworks, such as STSLR [7] and StepNet [61], have demonstrated strong performance by leveraging additional modalities or by incorporing large-scale pretaining using multimodal self-supervision signals (Tables rat14).

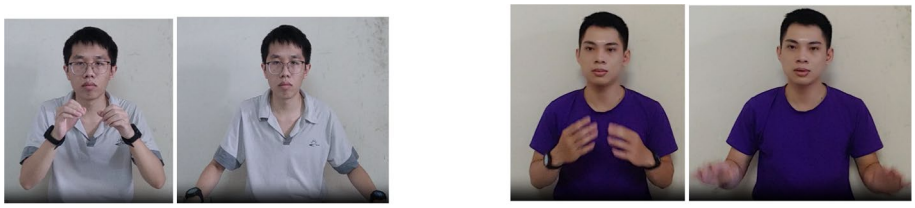
While our model has demonstrated strong performance on benchmarks such as WLASL100, WLASL300, and WLASL2000, as well as a custom 50-class Vietnamese Sign Language (VSL) dataset, we acknowledge that real-world Sign Language Recognition (SLR) systems must ultimately scale to vocabularies exceeding 10,000 signs. These datasets, while representative, remain limited in vocabulary size compared to practical deployment scenarios.

However, the consistent performance observed as we scale from WLASL100 (86.42%) to WLASL2000 (55.1%) suggests that the proposed CrossViVit architecture possesses a



(a) Eating (b) Comfortable (c) How old are you

Fig. 6 Example heatmap of sign actions



(a) Lost (b) Closing ceremony

Fig. 7 Example of signs with similar hand movement

degree of scalability and generalization capability. Moreover, the model maintains high top-10 accuracy on WLASL-100 (97.6%) and WLASL-300 (96.53%) using only RGB input-without relying on skeletal or optical flow modalities-which reinforces its practical viability.

(Q6): What challenges does our model need to address?

Before analyzing the error trend and challenge we meet in the SLR task only based on RGB, We have visualized the model's output based on the class token information from both branches to illustrate its ability to focus on relevant features. The Fig. 6 presents heatmaps for three distinct actions: eating, comforting, and how old are you. These visualizations demonstrate that the model effectively concentrates on hand movements while disregarding extraneous background information. This capability allows the model to rely solely on RGB information while achieving relatively high accuracy on large-scale datasets.

On the other hands, focusing on moving hand based on RGB make some challenges which is difficult to solve without additional hand skeleton features. Here, we observe that certain actions in Vietnamese Sign Language are highly similar, differing only in subtle hand movements, such as the degree of hand openness. As illustrated in the Fig. 7, the sign for (lost) differs from (closing ceremony) primarily in the extent to which the hand is open. This presents a significant challenge for sign language recognition models that rely solely on RGB-based feature extraction. Therefore, if real-time inference is not a strict requirement in practical deployment, incorporating additional information such as key points or skeleton data could enhance the model's ability to distinguish between these visually similar signs.

(Q7): How does our model perform on other fine-grained datasets? We have utilized the Sth-v2 dataset, which features an increased number of labels and samples, to thoroughly evaluate the model's performance on larger and more diverse datasets. This comprehensive assessment enables us to analyze the model's accuracy across various datasets. Moreover, the Sth-v2 dataset is categorized as fine-grained data, providing

Table 15 Performance comparison on Sth-v2

Model	Acc@1 (%)	Acc@5 (%)
TRN	48.8	77.6
SlowFast	61.7	
TimeSformer-HR	62.5	
TSM	63.4	88.5
ViVIT-L/16x2 FE	65.9	89.9
CrossViVit	***63.6***	***89.1***

essential insights into whether integrating information from the fine-grained branch in our methodologies results in superior performance compared to similar data types.

Table 15 demonstrates that CrossViVit performs relatively well on the Sth-v2 dataset, achieving an accuracy of 63.6%, higher than TRN [62], Slowfast [15], Times [63] and TSM [64]. While CrossVivit exhibits lower performance compared to ViVIT-L/16x2 FE, its computational efficiency is notably higher. Empirical evidence from both the Something-something and VSL datasets corroborates CrossVivit's adaptability to various datasets. The consistent performance on the Smth-v2 dataset is indispensable for real-world deployment of CrossVivit, given the vast vocabulary of sign languages.

5 Conclusion

In this paper, we introduced the CrossViViT model for Vietnamese Sign Language recognition, achieving a balance between computational cost and high accuracy. The CrossViViT architecture is designed to recognize sign language with high computational efficiency while maintaining accuracy. The model is divided into two main branches: (1) Coarse branch: This branch focuses on extracting general features from videos with low computational cost. The coarse branch uses multiple Transformer layers to process information from large-sized tokens. (2) Fine-grained branch: This branch focuses on extracting more detailed features from videos, complementing the information from the coarse branch. The fine-grained branch uses fewer Transformer layers and processes information from smaller-sized tokens. A unique feature of the CrossViViT architecture is the information exchange mechanism between the two branches through cross-attention. This mechanism allows the coarse branch to query information from the fine-grained branch, effectively aggregating information and improving the model's accuracy. However, several limitations remain. First, variations in signer identity, background, and lighting conditions can adversely affect generalization across datasets, such as the close similarity of actions, which misguide the accuracy of the model. Second, reliance on noisy depth input or low-resolution visual features may impede accurate recognition of signs that exhibit subtle differences in hand shape or motion. Thirdly, our model achieves strong recognition performance and competitive inference speed on high-end GPUs (e.g., RTX 4090 with 0.038 s/video), deploying it on edge devices remains a key challenge. The CrossViVit architecture, with its multi-branch Transformer design and high embedding dimension (448), is computationally demanding. While this complexity contributes to its accuracy, it may hinder real-time applications on mobile or embedded platforms. Addressing these challenges is essential for the model's successful widespread adoption in real-world applications.

For future research, we propose three promising directions to mitigate these issues: first, enhancing cross-branch information exchange mechanisms to improve both

efficiency and robustness; second, expanding the dataset to include a broader vocabulary and a more diverse set of signers to support better generalization; and finally, investigating the integration of natural language processing methodologies [65] to advance semantic understanding and disambiguation, particularly for signs that are visually similar but semantically distinct. Future work will expand the vocabulary coverage, explore continual and zero-shot learning, and apply model compression techniques to support lightweight deployment. These directions aim to bring us closer to practical, robust, and scalable SLR systems

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Data availability
Dataset will be available upon reasonable requestsThe data that support the findings of this study are available on request from the corresponding author Hoai Nam Vu (namvh@ptit.edu.vn).

Declarations

Ethics and consent to participate
Not applicable.

Consent to Publish
Not applicable.

Competing interests
The authors declare no competing interests.

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